MIL-C-48247 (MU)
15 September 1973

MILITARY SPECIFICATION

CARTRIDGES: 40MM, SMOKE, GROUND MARKER, XM713, XM714, XM715 AND XM716 METAL PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts for and loading, assembling and packing for Cartridges: 40MM, Smoke, Ground Marker, XM713, XM714, XM715 and XM716.

2. APPLICABLE DOCUMENTS

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition, General Specification for
MIL-C-5541	- Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys

STANDARDS

FEDERAL

FSC: 1310

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

FED-STD-151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for

MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Munitions Command

9277713 - Cartridge, 40MM, Red Smoke, Ground Marker, XM713

9277714 - Cartridge, 40MM, White Smoke, Ground Marker, XM714

9277715 - Cartridge, 40MM, Green Smoke, Ground Marker, XM715

9277716 - Cartridge, 40MM, Yellow Smoke, Ground Marker, XM716

8827889 - Gage, Chamber

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

. Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge.-The cartridge shall comply with all requirements specified as Drawing (Dwg.) 9277713, 9277714, 9277715, 9277716 (as applicable), all associated drawings and with all requirements specified in applicable specifications.

3.3 Transportation-Vibration.-The cartridge shall comply with the following requirements:

a. The cartridge shall be safe to transport following this test.

b. There shall be no evidence of external damage to the cartridge that will affect the intended function.

3.4 Functioning.-The cartridge shall comply with the following requirements:

a. Mean velocity of 247 ± 7 feet per second and a standard deviation not exceeding 4.5 feet per second.

b. No projectile shall stick in the gun bore.

c. Smoke duration time of 25 seconds.

d. Smoke ignition on impact or within 10 seconds of launch.

e. Fuze must function.

3.4.1 Hot temperature (160 degrees F. plus or minus 5 degrees F.).-The cartridge shall comply with the requirements of 3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

3.4.2 Ambient temperature (70 degrees F. plus or minus 5 degrees F.).-The cartridge shall comply with the requirements of 3.4.

3.4.3 Cold temperature (-65 degrees F. plus or minus 5 degrees F.).-The cartridge shall comply with the requirements of 3.4.

3.5 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.5)

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements when such material is purchased by the contractor or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1.) The first article sample shall consist of the following:

a. Twenty-five (25) bodies (Dwg. 8277717).
b. Twenty-five (25) bases (Dwg. 9277718).
c. Three (3) test panels or parts of body for salt spray (Dwg. 9277717).
d. Twenty-five (25) projectile assemblies (Dwg. 9277725, 9277726, 9277727, or 9277728, as applicable).
e. Twenty-five (25) cartridges (Dwg. 9277713, 9277714, 9277715 or 9277716 as applicable).
f. Four-hundred and forty (440) cartridges to a Government proving ground for functioning tests (Dwg. 9277713, 9277714, 9277715, or 9277716 as applicable).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

All parts and completed assemblies submitted shall have been produced by the Contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process (including heat treatment procedures and equipment, material used, drawing specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All parts and assemblies shall be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawing.

4.2.3 Rejection.-If any part or assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any part or assembly. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4 Remaining first article inspection samples.-Unless otherwise specified by the Contracting Officer, loaded cartridges shall not be submitted for proving ground tests until the inert components and sub-assemblies submitted in accordance with 4.2 have been accepted.

6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

4.2.4.1 Prior to submission.-Prior to submission of the four-hundred and forty (440) cartridges to a Government proving ground for functioning tests, the 440 cartridges shall be inspected at the contractor's plant for the following tests:

4.2.4.1.1 Transportation-Vibration (See 3.3).-The 440 cartridges packed in their shipping containers shall be subjected to this test.

4.2.4.1.2 Leak test (See Dwg. 9277713, 9277714, 9277715 or 9277716 as applicable).-After completion of test specified in 4.2.4.1.1, the 440 cartridges shall be removed from their regular shipping containers and subjected to this test.

4.2.5 Rejection.-If any assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly in the sample to comply with any of the state requirements. In the event of rejection, the Government reserves the right to require the contractor to take correxitive action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.6 Proving ground tests.-The cartridges shall be subjected by the Government to the following tests:

a. Functioning at hot temperature (see 3.4.1.-Eighty (80) cartridges shall be subjected to this test.

b. Functioning at ambient temperature (see 3.4.2).-Two-hundred and eighty (280) cartridges shall be subjected to this test.

c. Functioning at cold temperature (see 3.4.3).-Eighty (80) cartridges shall be subjected to this test.

4.2.6.1 Rejection.-The first article sample shall be rejected if:

a. Mean velocity or standard deviation fail the applicable requirement (3.4, a).

b. Any projectile sticks in the gun bore in any phase (3.4, b).

c. A total of eight (8) or more fail any of the following in the hot phase:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

(1) Smoke duration time (3.4,c).
(2) Smoke ignition time (3.4,d).
(3) Fuze must function (3.4,e).
(4) Incorrect smoke color.

d. A total of twenty-two (22) or more in the ambient phase fail any of the following:
(1) Smoke duration time (3.4,c).
(2) Smoke ignition time (3.4,d).
(3) Fuze must function (3.4.,e).
(4) Incorrect smoke color.

e. A total of eight (8) or more in the cold phase fail any of the following:
(1) Smoke duration time (3.4,c).
(2) Smoke ignition time (3.4,d).
(3) Fuze must function (3.4,e).
(4) Incorrect smoke color.

f. Any premature smoke ignition in gun bore in any phase.

In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity. Until the first article quantity is accepted, the contractor is in no way authorized by the Government to proceed with regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.-

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Change to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Cartridge case loaded assemblies from one interfix lot number from one manufacturer.
b. Projectile assemblies from one interfix lot number from one manufacturer.
c. Fuze assemblies from one interfix lot number from one manufacturer.
d. Ignition mix from one interfix lot number from one manufacturer.
e. Smoke composition produced by one manufacturer under one continuous set of operating conditions and which consist of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plan for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Individual defects shall have an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Body (See Dwg. 9277717).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

Major			
101.	Largest outside diameter	Gage	01001
102.	Concentricity of largest outside diameter with bourrelet diameter	Gage	01002
103.	Bourrelet diameter	Gage	01003
104.	Concentricity of body diameter rear of rotating band with bourrelet diameter	Gage	01004
105.	Concentricity of body diameter forward of rotating band with bourrelet diameter	Gage	01005
106.	Concentricity of hole with bourrelet diameter	Gage	01006
107.	Concentricity of base counterbore with bourrelet diameter	Gage	01007
108.	Depth of base counterbore	Gage	01008
Minor			
201.	Body diameter rear of rotating band	Gage	01009
202.	Body diameter forward of rotating band	Gage	01010
203.	Base cavity diameter	Gage	01011
204.	Seal cavity diameter	Gage	01012
205.	Total length	Gage	01013
206.	Distance to rotating band	Gage	01014
207.	Width of rotating bands	Gage	01015
208.	Diameter between rotating bands	Gage	01016
209.	Evidence of poor workmanship (see 3.5)	Visual	01017

4.3.2.2 Base (see Dwg. 9277718).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Concentricity of fuze cavity with large outside diameter	Gage	02001

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

102.	Concentricity of hole with large outside diameter	Gage	02002
103.	Concentricity of second largest outside diameter with largest outside diameter	Gage	02003
Minor			
201.	Total height	Gage	02004
202.	Large outside diameter	Gage	02005
203.	Evidence of poor workmanship (see 3.5)	Visual	02006

4.3.2.3 Projectile, assembly, smoke (see Dwg. 9277725, 9277726, 9277727, or 9277728 as applicable).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Weight	Balance	03001
102.	Total length	Gage	03002
Minor			
201.	Evidence of poor workmanship (see 3.5)	Visual	03003

4.3.2.4 Cartridge (see Dwg. 9277714, 9277715, 9277716 or 9277717 as applicable).-

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Chamber gage failure (after complete cure)	8827889	04001
2.	Incorrect paint identification color	Visual	04002
Major			
101.	Marking unidentifiable	Visual	04003

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

Minor			
201.	Total length	Gage	04004
202.	Evidence of poor workmanship (see 3.5)	Visual	04005

4.3.2.5 Unsealed metal box (see Dwg. 9209204).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Incorrect number of cartridges	Visual	05001
102.	Desiccant bags missing	Visual	05002
103.	Rubber gasket damaged or improperly assembled	Visual	05003
Minor			
201.	Fillers missing	Visual	05004
202.	Tube missing	Visual	05005

4.3.2.6 Sealed metal box (see Dwg. 9209204).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor			
201.	Marking misleading or unidentifiable	Visual	06001
202.	Contents loose	Manual	06002

4.3.2.7 Sealed wooden packing box (see Dwg. 9209205).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

Minor			
201.	Marking misleading or unidentifiable	Visual	07001
202.	Contents loose	Manual	07002
203.	ICC nomenclature missing	Visual	07003

4.3.3 Testing.-

4.3.3.1 Static burn time of projectile assembly (See Dwg. 9277725, 9277726, 9277727 or 9277728 as applicable), Major Defect, Code No. 08001.-Thirty-two (32) projectile assemblies shall be selected from each lot and subjected to this test. If two (2) or more assemblies fail to comply with the applicable drawing requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2 Salt spray of body (See Dwg. 9277717), Major Defect, Code No. 09001.-Sampling and rejection shall be in accordance with MIL-C-5541. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.3 Security of projectile assembly (See Dwg. 9277713, 9277714, 9277715 or 9277716 as applicable), Major Defect, Code No. 10001.-Fifty (50) cartridges shall be selected from each lot and subjected to this test. If four (4) or more assemblies fail to comply with the applicable drawing requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.4 Leak test of cartridge (see Dwg. 9277713, 9277714, 9277715 or 9277716 as applicable), Major defect, Code No. 11001.-This test shall be conducted 100 percent. Any cartridge which fails to comply with the applicable drawing requirement shall be classified defective and removed from the lot. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.5 Transporation-vibration (see 3.3), Major defect, Code No. 12001.-One-hundred and thirty-two (132) cartridges shall be selected from each lot, packed in their regular shipping containers and subjected to this test. The cartridges shall be observed and examined visually without

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

disassembly for any evidence of failure to comply with the requirements as classified in Table I. The lot shall be rejected if any defect as classified in Table I occurs. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4.

TABLE I

Defect	Classification	Code No.
Cartridge not safe to transport following test (see 3.3.a)	Critical	12002
Cartridge damaged after test (see 3.3.b)	Major	12003

4.3.3.6 Leak test of cartridge (see dwg. 9277713, 9277714, 9277715 or 9277716 as applicable), Major defect. Code No. 13001.-After completion of test specified in 4.3.3.5, the one-hundred and thirty-two (132) samples shall be removed from their regular shipping containers and subjected to this test. If eight (8) or more samples fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. Samples which fail this test shall be discarded and replaced with units which have been subjected to test specified in 4.3.3.4 prior to subsequent testing. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.7 Functioning (See 3.4), Major defect, Code No. 14001.-After completion of test specified in 4.3.3.6, the one-hundred and thirty-two (132) cartridges shall be subjected to this test.

4.3.3.7.1 Hot temperature (see 3.4.1).-Twenty-six (26) cartridges shall be subjected to this test.

4.3.3.7.2 Ambient temperature (see 3.4.2).-Eighty (80) cartridges shall be subjected to this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

4.3.3.7.3 Cold temperature (see 3.4.3).-Twenty-six (26) cartridges shall be subjected to this test.

4.3.3.7.4 Rejection.-The lot shall be rejected if:

a. Mean velocity or standard deviation fail the applicable requirement (3.4.a).

b. Any projectile sticks in the gun bore in any phase (3.4.b).

c. A total of four (4) or more in the hot phase fail any of the following:

(1) Smoke duration time (3.4.c).
(2) Smoke ignition time (3.4.d).
(3) Fuze must function (3.4.e).
(4) Incorrect smoke color.

d. A total of eight (8) or more in the ambient phase fail any of the following:

(1) Smoke duration time (3.4.c).
(2) Smoke ignition time (3.4.d).
(3) Fuze must function (3.4.e).
(4) Incorrect smoke color.

e. A total of four (4) or more in the cold phase fail any of the following:

(1) Smoke duration time (3.4.c).
(2) Smoke ignition time (3.4.d).
(3) Fuze must function (3.4.e).
(4) Incorrect smoke color.

f. Any premature smoke ignition in gun bore in any phase.

4.3.3.8 Suitability test plan (Informational test only), Code No. 14001.-Beginning with the first lot produced by each producer and continuing by each producer and continuing for three (3) accepted lots (lots not accepted by disposition) that are accepted in accordance with 4.3.3.7, an extra sample of eighty-eight (88) rounds shall be selected from each lot and preconditioned and tested as follows:

a. The eighty-eight (88) rounds shall be packaged in their regular shipping containers and subjected to transporation vibration environmental in accordance with MIL-STD-331, Test No. 104.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

b. After transportation, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

c. After temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

30 rounds	+ 165 degrees F. ± 5 degrees F.
28 rounds	ambient (between ∓40 and +110 degrees F.)
30 rounds	-65 degrees F. ± 5 degrees F.

d. Fire the rounds in accordance with Paragraph 4.4.6 . and with TECOM Material Test Procedure, TECP 700-700 Volume 4. Test results, failure analysis reports and other information as requested shall be forwarded to Picatinny Arsenal, Attn: SARPA-QA-A-S for evaluation.

4.3.3.9 Air pressure test of packed seal metal box (See Dwg. 9209204), Major defect, Code No. 17001.-One (1) packed sealed metal box shall be selected from each twenty (20) boxes packed. If the metal box fails to comply with the applicable drawing, the quantity represented by the sample shall be rejected. The test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and in 4.3.3 (See 6.3).

4.4 Test methods and procedures.-

4.4.1 Static burn time of projectile assembly.-This test shall be conducted in accordance with a Government approved procedure.

4.4.2 Salt spray of body.-This test shall be conducted in accordance with FED-STD-151, Method 811.1. Specimens so tested shall not be returned to the lot. Salt solution shall be 5 percent.

4.4.3 Security of projectile assembly.-The cartridge shall be placed in a Government approved fixture and the axial farce specified on Dwg. 9277713, 9277714, 9277715 or

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

9277716 as applicable shall be applied. Assembly shall be pulled until total separation occurs and data shall be recorded. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.4 Leak test of cartridge.-The cartridge shall be placed in a Government approved fixture and the requirement specified on the applicable drawing shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.5 Transporation vibration.-The cartridges shall be packaged and packed in accordance with dwgs. 9209204, and 9209205 and tested in accordance with Test No. 104 of MIL-STD-331, except that each box shall be vibrated at the specified amplitudes for four (4) hours in each of three (3) different positions (i.e., box positioned so that the cartridges are vertical with case end down, box positioned so that the cartridges are horizontal, and box positioned so that the cartridges are vertical with case end up. Aftem the test, the box packing and the cartridges shall be examined to determine compliance with the requirements. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.6 Functioning.-This test shall be performed at a Government-owned proving ground. Hot, ambient and cold temperature. The rounds being tested shall be temperature conditioned at least eight (8) hours, then fired within fifteen (15) minutes after being removed from the temperature soak. Gravel: The cartridge shall be fired to impact on a six inch thick loose gravel pad. The gravel pad used shall have a 1/2 inch maximum stone size. Steel: The cartridge shall be fired to impact against a vertical 2 1/2 inch thick sheet of mild steel plate. (proper safety precautions should be exercised to protect all exposed personnel).

The velocity shall be measured at twenty (20) feet from the muzzle of the launcher, using two sets of screens, each set of screens having an electronic counter. One set of screens shall be placed 14 and 26 feet from the launcher and the other set 15 and 25 feet from the launcher. Note:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

To insure that the choronographs are read properly. The ratio of the two times shall not differ from 1.200 by more than plus or minus 0.005. If the ratio differs from 1.200 by more than plus or minus 0.005 the velocity results for that round shall be discarded and excluded from the calculation of mean velocity and velocity standard deviation (no more than eight (8) rounds in a sample size of 132 may be disregarded). If the applicable number of improper ratios is exceeded additional rounds must be fired. Observation shall be made and results recorded for functioning, mean velocity and standard deviation. The average of all rounds fired per sample (except above noted improper ratio(s) shall meet the requirements of 3.4.a.

4.4.6.1 Test Validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.4.7 Air pressure of packed, sealed metal box.-The metal box shall be placed in a Government approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Observation shall be made for leakage. This test is a non-destructive test. Boxes so tested may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with Dwg. 9209204.

5.2 Packing.-

5.2.1 Level A.-The cartridge preserved and packaged in accordance with 5.1.1, shall be packed in accordance with Dwg. 9209205.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

5.3 Marking.-Marking shall be in accordance with Dwgs. 9209204, 9209205, and Code of Federal Regulations, Title 49, Parts 0-190.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types: Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirement Lists (DD Form 1423).

The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabrication or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

6.4 Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the U.S. Army Munitions Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S.

6.5 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S.

6.6 Standard deviation shall be calculated from the following formula:

$$S = \sqrt{\frac{(X - \bar{X})^2}{n - 1}}$$

where:

X = Each individual value
$\bar{X}$ = Sample arithmetic mean $\frac{\Sigma X}{n}$

n = Sample size
$\Sigma(X - \bar{X})^2$ = The sum of the squares of the differences between the sample mean and each individual values.
$n - 1$ = One less than the sample.

6.6.1 For a faster and easier method of arriving at the standard deviation, the following alternate method may be used:

$$S = \sqrt{\frac{n(\Sigma x^2) - (\Sigma X)^2}{n \quad (n-1)}}$$

6.7 Proving ground test summary:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

Test	Sample Size	Requirement
Functioning		
(1) First article		
a. Hot	80	4.2.6.1
b. Ambient	280	4.2.6.1
c. Cold	80	4.2.6.1
(2) Production		
a. Hot	26	4.3.3.7.4
b. Ambient	80	4.3.3.7.4
c. Cold	26	4.3.3.7.4

6.8 Distribution of ammunition data cards.-Distribution of data cards shall include the following: Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S, Dover, New Jersey 07801.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1310-A035

22

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.